



Curriculum Vitae

Jiandong Tian, Ph.D., Professor

Robot/Computer Vision Group

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Professional Background

Education

2005.9- 2011.7 Ph.D., Pattern Recognition and Intelligent System.

Shenyang Institute of Automation, Chinese Academy of Sciences

2001.9-2005.7 B.S. degree, Automation, Heilongjiang University, China

Employment

2017.1-Present Full professor, Shenyang Institute of Automation, Chinese Academy of Sciences

2015.6-2016.6 Visiting scholar, Center for Visual Computing, University of California, San Diego (UCSD)

2013.1-2016.12 Associate professor, Shenyang Institute of Automation, Chinese Academy of Sciences

2011.7-2013.1 Assistant professor, Shenyang Institute of Automation, Chinese Academy of Sciences

Research Interests: -My research interests focus on robot/computer vision, especially in complicated illumination and meteorological conditions.

Scopes includes,

1. **Vision in Complex Illumination.** Outdoor illumination and reflectance modelling; Shadow and highlight processing.

2. **Vision in bad weather.** Desnowing and deraining; fog/scattering modelling and removal; underwater descattering.
3. **High-level vision tasks.** Tracking in 2D/3D; Object Detection with Domain Adaptation; VLA and Word model for robots.
4. **Cameras for computer vision.** Designing a shadow-free, weather-free, and highlight-free camera; Under water camera; RGB-NIR camera; Polarization camera; HDR/Multi-modal/TOF cameras; SWIR camera.

Selected Publications (* means corresponding author paper)

1. Baojie Fan, Xiaotian Li, Yuhan Zhou, Caixia Xia, Huijie Fan, Fengyu Xu, and **Jiandong Tian***, "MGAF: LiDAR-Camera 3D Object Detection with Multiple Guidance and Adaptive Fusion", *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2025.
2. Zhenshuo Yang, Chunhui Hao, Yang Lu, Yiming Su, Yukuan Zhang, Junchao Zhang, and **Jiandong Tian***, "A Novel Dehazing Approach: Recovery of Color and Polarization Information Using Polarized Characteristics", *IEEE Transactions on Multimedia*, 2025.
3. Shijun Zhou, Yiming Su, Dongyue Wang, Weihong Ren, and **Jiandong Tian***, "LFUID: Light Field-Based Underwater Image Formation, Restoration, and Real-World Dataset", *IEEE Transactions on Industrial Informatics*, 2025.
4. Qian He, Wenqi Liang, Chunhui Hao, Gan Sun, and **Jiandong Tian***, "GLAM: Global-Local Variation Awareness in Mamba-based World Model", *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI)*, 2025. (Oral)
5. Yang Lu, **Jiandong Tian***, Yiming Su, Jiawei Liu, Chunhui Hao, Chenghai Yue, and Xin Yang, "Polarization-Aware Low-Light Image Enhancement for Nighttime Intelligent Vehicles", *IEEE Transactions on Intelligent Vehicles*, 2025.
6. **Jiandong Tian***, Shijun Zhou, Baojie Fan, and Hui Zhang, "A Novel Image Formation Model for Descattering", *IEEE Transactions on Pattern Analysis and Machine*

Intelligence, 2024.

7. Yajing Liu, Shijun Zhou, Xiyao Liu, Chunhui Hao, Baojie Fan, and **Jiandong Tian***, "Unbiased Faster R-CNN for Single-source Domain Generalized Object Detection", *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024. **(Highlight Paper)**
8. Xiaotian Li, Baojie Fan*, **Jiandong Tian**, and Huijie Fan, "GAFusion: Adaptive Fusing LiDAR and Camera with Multiple Guidance for 3D Object Detection", *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024.
9. Yang Lu, **Jiandong Tian***, Yiming Su, Yidong Luo, Junchao Zhang, and Chunhui Hao, "A Hybrid Polarization Image Demosaicking Algorithm Based on Inter-Channel Correlation", *IEEE Transactions on Computational Imaging*, 2024.
10. Shijun Zhou, Zhen Zhang, Yajing Liu, and **Jiandong Tian***, "Restoration on High Turbidity Water Images under near-Field Illumination Using a Light-Field Camera", *IEEE Transactions on Computational Imaging*, 2024.
11. Yiming Su, Zhen Zhang, Yang Lu, Shibin Liu, Yandong Tang, and **Jiandong Tian***, "Underwater Descattering with Polarization Line Constraint", *Optics and Lasers in Engineering*, 2024.
12. Yang Lu, Weihong Ren, Yiming Su, Zhen Zhang, Junchao Zhang, and **Jiandong Tian***, "Polarization Image Demosaicking Based on Homogeneity Space", *Optics and Lasers in Engineering*, 2024.
13. Ren, Weihong*, Jinguo Luo, Weibo Jiang, Liangqiong Qu, Zhi Han, **Jiandong Tian**, and Honghai Liu, "Learning Self-and Cross-Triplet Context Clues for Human-Object Interaction Detection", *IEEE Transactions on Circuits and Systems for Video Technology*, 2024.
14. Pei, Wenjie, Tongqi Xia, Fanglin Chen, Jinsong Li, **Jiandong Tian**, and Guangming Lu*, "Sa²vp: Spatially Aligned-and-Adapted Visual Prompt", *the Proceedings of the AAAI Conference on Artificial Intelligence (AAAI)*, 2024.
15. Jingwen Zhang, Zikun Zhou*, Guangming Lu, **Jiandong Tian**, and Wenjie Pei*, "Robust 3d Tracking with Quality-Aware Shape Completion", *the Proceedings of the AAAI Conference on Artificial Intelligence (AAAI)*, 2024.

16. Baojie Fan, Wuyang Zhou, Yushi Yang, and **Jiandong Tian***, "Integrating Scaling Strategy and Central Guided Voting for 3d Point Cloud Object Tracking", *IEEE Robotics and Automation Letters*, 2024.
17. Baojie Fan, Kexin Chen, Guoping Jiang, and **Jiandong Tian***, "Two-Way Complementary Tracking Guidance", *IEEE Transactions on Circuits and Systems for Video Technology*, 2023.
18. Baojie Fan, Kexin Zhang, and **Jiandong Tian**, "HCPVF: Hierarchical Cascaded Point-Voxel Fusion for 3d Object Detection", *IEEE Transactions on Circuits and Systems for Video Technology*, 2023.
19. Jiawei Liu, Qiang Wang, Huijie Fan, **Jiandong Tian**, and Yandong Tang, "A Shadow Imaging Bilinear Model and Three-Branch Residual Network for Shadow Removal", *IEEE Transactions on Neural Networks and Learning Systems*, 2023.
20. Baojie Fan, Hui Zhang, Yang Cong, Yandong Tang, Huijie Fan, and **Jiandong Tian***, "Dual Aligned Siamese Dense Regression Tracker", *IEEE Transactions on Image Processing*, 2022.
21. Shuang Wu, Wenjie Pei*, Dianwen Mei, Fanglin Chen, **Jiandong Tian**, and Guangming Lu*. "Multi-Faceted Distillation of Base-Novel Commonality for Few-Shot Object Detection", *the European Conference on Computer Vision (ECCV)*, 2022.
22. Wenjie Pei, Shuang Wu, Dianwen Mei, Fanglin Chen, **Jiandong Tian**, and Guangming Lu, "Few-Shot Object Detection by Knowledge Distillation Using Bag-of-Visual-Words Representations", *the European Conference on Computer Vision (ECCV)*, 2022.
23. Weihong Ren, Xinchao Wang, **Jiandong Tian**, Yandong Tang, and Antoni B Chan, "Tracking-by-Counting: Using Network Flows on Crowd Density Maps for Tracking Multiple Targets", *IEEE Transactions on Image Processing*, 2020.
24. Baojie Fan, Yang Cong, Yandong Tang, **Jiandong Tian**, and Chenliang Xu, "Structured and Consistent Multi-Layer Multi-Kernel Subtask Correction Filter Tracker", *IEEE Transactions on Circuits and Systems for Video Technology*, 2020.
25. Baojie Fan, Yang Cong, **Jiandong Tian***, and Yandong Tang, "Dynamic and reliable subtask tracker with general Schatten p-norm regularization", *Pattern Recognition*, 2021.
26. Baojie Fan, **Jiandong Tian**, Yan Peng, and Yandong Tang, "Discriminative Siamese

- Complementary Tracker with Flexible Update", *IEEE Transactions on Multimedia*, 2021.
27. Pengyue Li, **Jiandong Tian***, Yandong Tang, Guolin Wang, and Chengdong Wu, "Deep Retinex Network for Single Image Dehazing", *IEEE Transactions on Image Processing*, 2020.
 28. Weihong Ren, Xinchao Wang, **Jiandong Tian**, Yandong Tang, and Antoni B Chan, "Tracking-by-counting: Using network flows on crowd density maps for tracking multiple targets", *IEEE Transactions on Image Processing*, 2020.
 29. Baojie Fan, Wei Chen, Yang Cong, and **Jiandong Tian**, "Dual refinement underwater object detection network", *the European Conference on Computer Vision (ECCV)*, 2020.
 30. Baojie Fan, Yang Cong, **Jiandong Tian**, and Yandong Tang, "Reliable Multi-Kernel Subtask Graph Correlation Tracker", *IEEE Transactions on Image Processing*, 2020.
 31. Fan Baojie, Yang Cong, Yandong Tang, **Jiandong Tian**, and Chenliang Xu, "Structured and Consistent Multi-Layer Multi-Kernel Subtask Correction Filter Tracker", *IEEE Transactions on Circuits and Systems for Video Technology*, 2020.
 32. **Jiandong Tian**, Zhi Han, Weihong Ren, and Yandong Tang, "Snowflake removal for videos by foreground and background decomposition via low rank factorization", *IEEE Transactions on Multimedia*, 2018.
 33. **Jiandong Tian**, Zak Murez, Tong Cui, Zhen Zhang, David Kriegman, and Ravi Ramamoorthi, "Depth and Image Restoration from Light Field in a Scattering Medium", *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2017.
 34. Weihong Ren, **Jiandong Tian***, Zhi Han, and Yandong Tang, "Video Desnowing and Deraining Based on Matrix Decomposition", *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2017.
 35. Liangqiong Qu, **Jiandong Tian***, Shengfeng He, Yandong Tang, and Rynson Lau, "DeshadowNet: A Multi-context Embedding Deep Network for Shadow Removal", *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2017.
 36. Weihong Ren, **Jiandong Tian***, Yandong Tang, "Specular Reflection Separation with Color-Lines Constraint", *IEEE Transactions on Image Processing*, 2017.

37. Liangqiong Qu, Shengfeng He, Jiawei Zhang, **Jiandong Tian**, Yandong Tang, and Qingxiong Yang, "RGBD Salient Object Detection via Deep Fusion", *IEEE Transactions on Image Processing*, 2017.
38. Zhi Han, **Jiandong Tian**, Liangqiong Qu, and Yandong Tang, "A New Intrinsic-Lighting Color Space for Daytime Outdoor Images", *IEEE Transactions on Image Processing*, 2017.
39. **Jiandong Tian**, Zhigang Duan, Weihong Ren, Zhi Han, Yandong Tang, "Simple and Effective Calculations about Spectral Power Distributions of Outdoor Light Sources for Computer Vision", *Optics Express*, 2016.
40. **Jiandong Tian***, Xiaojun Qi, Liangqiong Qu, and Yandong Tang, "New Spectrum Ratio Properties and Features for Shadow Detection", *Pattern Recognition*, 2016.
41. **Jiandong Tian** and Yandong Tang, "Wavelength-sensitive-function controlled reflectance reconstruction", *Optics Letters*, 2013.
42. **Jiandong Tian** and Yandong Tang, "Linearity of each channel pixel values from a surface in and out of shadows and its applications", *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2011.
43. **Jiandong Tian**, Jing Sun, and Yandong Tang, "Tricolor attenuation model for shadow detection", *IEEE Transactions on image processing*, 2009.

Research Grant (Role in project)

1. 2024.12-2027.11:"China-Lebanon Joint Laboratory for Intelligent Robots", (Grant Number: 2024YFE0115500), **Principal Investigator**, National Key R&D Program of China, RMB: 400,000,0.
2. 2020.01-2022.12:"Research and development of emergency robot equipment inside dangerous buildings at disaster sites", (Grant Number: SQ2019YFC150037), **Principal Investigator**, National Key R&D Program of China, RMB:176,500,00.
3. 2021.01-2024.12:"Vision-based robot sensing and targets recognition in complex outdoor conditions", (Grant Number: U2013210), **Principal Investigator**, Key Program of National Natural Science Foundation of China, RMB: 279,000,0.

4. 2019.01-2021.12:” Multi-modal Robotic Visual Perception System”, (Grant Number: 2019000065), **Principal Investigator**, Project for Research Instrument Development, RMB: 359,000,0.
5. 2018.01-2021.12:”All weather computer vision”, (Grant Number: Y8A1065101), **Principal Investigator**, Chinese Academy of Sciences, RMB: 180, 000,0.
6. 2023.01-2023.12:” Robot visual perception in complex environments”, (Grant Number: 2023JH6/100500005), **Principal Investigator**, the LiaoNing Science Fund for Distinguished Young Scholars, RMB: 100, 000,0.
7. 2017.01-2020.12:” Multimode perception system for Tri-Co robots on complicated illuminations and weather”, (Grant Number: 91648118), **Principal Investigator**, National Natural Science Foundation of China, RMB: 680,000.
8. 2015.01-2018.12:” Illumination modeling and its invariant algorithms for computer vision”, (Grant Number: 61473280), **Principal Investigator**, National Natural Science Foundation of China, RMB: 800,000.
9. 2016.1-2018.12: “Deep learning and illumination processing for robots”, **Principal Investigator**, Siasun Robot & Automation Co.,Ltd., RMB: 800,000.
10. 2013.12-2016.12:” Illumination, Reflectance, and camera modeling”, (Grant Number: Y3A1065101), **Principal Investigator**, Chinese Academy of Sciences. RMB: 400, 000.
11. 2011.01-2013.12:” Physical and imaging properties of outdoor shadow and its removal methods”, (Grant Number: 61102116), **Principal Investigator**, National Natural Science Foundation of China, RMB: 280,000.

Honors and Awards

- [1] 2024, Scientific Leadership Award, The 5th Annual Robotics Industry Conference.
- [2] 2024, CAA Young Scientist Award, Chinese Association of Automation (CAA).
- [3] 2024, Second Prize, Natural Science Award, Chinese Association of Automation (CAA).
- [4] 2024, Second Prize, Natural Science Award, Chinese Society for Optical Engineering (CSOE).
- [5] 2020, First Prize (Team Leader), The 1st Navy Underwater Unmanned Vehicle (UUV) Field Competition (Target Recognition Group).

[6] 2020, First Prize of Liaoning Provincial Natural Science Award.

[7] 2019, First Prize, Natural Science Award, Chinese Association of Automation (CAA).

[8] 2018, Outstanding Youth Innovation Promotion Award, Chinese Academy of Sciences.

[9] 2016, First prize, Academic award of nature science in Liaoning province.

Professional Service

[1] Expert Panelist: National "Natural Disaster Prevention and Control Technical Equipment Modernization Project" (R&D); designated expert for the Intelligent Unmanned Emergency Rescue sector.

[2] Guideline Compilation Expert: National Key R&D Program: "Major Natural Disaster Prevention and Control and Public Safety."

[3] Director: China-Lebanon Joint Laboratory for Intelligent Robots.

[4] Assistant to the Director: National Key Laboratory of Robotics and Intelligent Systems.

[5] Deputy Director: Liaoning Provincial Key Laboratory of Machine Intelligence.

[6] Vice Chairman: Technical Committee on Simulation Technology Application, China Simulation Federation.

[7] Standing Committee Member: Technical Committee on Co-existing Robots, Chinese Association of Automation.

[8] Council Member, Technical Committee on Intelligent Industrial Data Analytics and Optimization, Operations Research Society of China.

[9] IEEE, Senior member.

[10] Associate Editor: IEEE Signal Processing Letters

[11] Guest Editor: Journal of Imaging, Electronics, Sensors, Frontiers in Robotics and AI, Frontiers in Computer Science.

[12] Reviewer: IEEE Transactions on Pattern Analysis and Machine Intelligence, IEEE Transactions on Image Processing, Pattern Recognition, IEEE Signal Processing Letters, IEEE Transactions on Industrial Informatics, IEEE Transactions on Systems, Man, and Cybernetics, Journal of the Optical Society of America A, Applied Optics, Computers & Graphics, and Remote Sensing, IEEE CVPR, IEEE ICCV, NeurIPS, AAAI.

Advisees

[1] Current Ph.D. students

Shijun Zhou, Yajing Liu, Mingxue Li, Zhenshuo Yang, Qian He, Dongyue Wang, Lanqi Li.

[2] Alumni

Dr. Yang Lu (Shenyang Aircraft Corporation)

Dr. Yiming Su (BYD Co., Ltd)

Dr. Zhen Zhang (BYD Co., Ltd)

Dr. Denglu Wu (Research Scientist, National University of Singapore)

Dr. Zhigang Duan (Inspur Group Co., Ltd)

Dr. Liangqiong Qu (Assistant Professor, The University of Hong Kong)

Dr. Weihong Ren (Assistant Professor, Harbin Institute of Technology, ShenZhen, China)

References

Available upon request.